

RAG-Based Moodle Plugin: Practical Example

This document provides a simple, presentation-friendly example of how a Retrieval-Augmented Generation (RAG) system can use Moodle course content to answer student questions.

1. Example Moodle Content

Assume a Moodle course contains a PDF named 'Introduction_to_Cloud_Computing.pdf' with the following content:

- Chapter 1: Cloud Computing — Cloud computing is the delivery of computing resources such as servers, storage, databases, networking, and software over the internet.
- Chapter 2: Cloud Service Models — The three major cloud service models are Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS).
- Chapter 3: Cloud Deployment Models — Cloud deployment models include public cloud, private cloud, hybrid cloud, and community cloud.

2. Chunking – Breaking the Document into Smaller Pieces

Instead of sending the entire PDF to the AI model, the RAG system divides the content into smaller meaningful sections called chunks.

Chunk 1: Cloud computing is the delivery of computing resources such as servers, storage, databases, networking, and software over the internet.

Chunk 2: The three major cloud service models are Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS).

Chunk 3: Cloud deployment models include public cloud, private cloud, hybrid cloud, and community cloud.

Simple concept: Large PDF → Smaller meaningful chunks.

3. Embedding – Converting Text into Numbers

Each chunk is passed through an embedding model. The model converts the meaning of the text into a numerical vector. The actual vector can contain hundreds or thousands of dimensions, depending on the embedding model.

Example:

Chunk 1: "Cloud computing is the delivery of computing resources over the internet."

Conceptual embedding: [0.21, -0.45, 0.73, 0.18, -0.62, ...]

Chunk 2: "The three major cloud service models are IaaS, PaaS and SaaS."

Conceptual embedding: [0.31, -0.41, 0.69, 0.22, -0.57, ...]

The important point is that embeddings represent semantic meaning. Therefore, a question such as 'What are the different types of cloud services?' can be matched with content about IaaS, PaaS, and SaaS even when the wording is not exactly the same.

4. Vector Database – Storing and Searching the Embeddings

The generated vectors, text chunks, and useful Moodle metadata are stored in a vector database such as FAISS, Chroma, Qdrant, or another vector store.

Chunk ID	Text	Embedding	Metadata
001	Cloud computing is...	[0.21, -0.45, ...]	Course: Cloud Computing
002	Three major service models...	[0.31, -0.41, ...]	Chapter: 2
003	Deployment models include...	[0.15, -0.32, ...]	Chapter: 3

Example Moodle metadata can include:

- course_id = 105
- course_name = Cloud Computing
- section = Chapter 2
- resource = Cloud_Computing.pdf
- chunk_id = 002

Metadata is important because the Moodle plugin can use it to identify the source of the information and apply access restrictions so that students retrieve only content they are authorized to access.

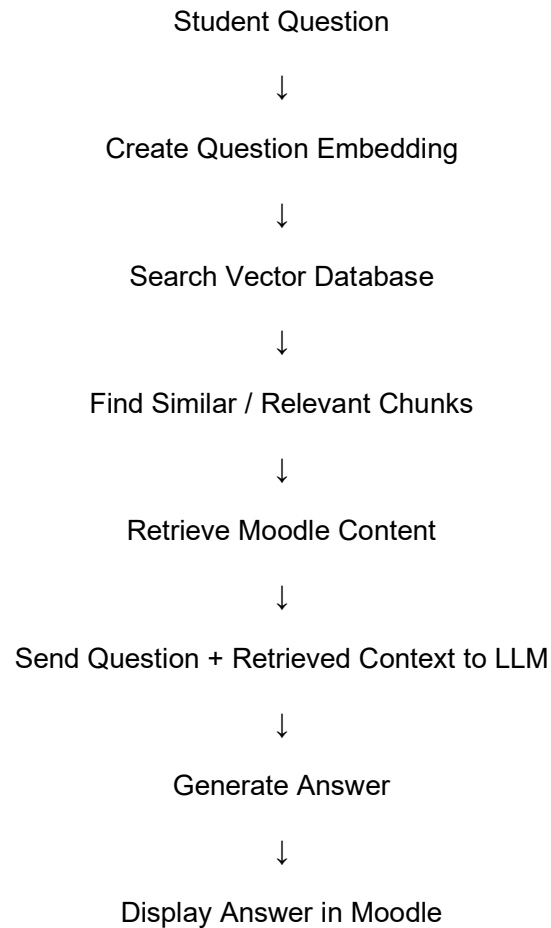
5. What Happens When a Student Asks a Question?

Student question: "What are the three cloud service models?"

1. The student submits the question through Moodle.
2. The question is converted into an embedding using the same or compatible embedding approach.
3. The system performs a similarity search in the vector database.
4. The system identifies the most relevant chunks. In this example, Chunk 002 is highly relevant.
5. The retrieved Moodle content is combined with the student's question.

6. The combined context is sent to the Large Language Model (LLM).
7. The LLM generates a grounded answer.
8. The Moodle plugin displays the answer and can optionally show the source Moodle resource.

6. Complete RAG Flow



7. Example Final Answer

The three major cloud service models are IaaS (Infrastructure as a Service), PaaS (Platform as a Service), and SaaS (Software as a Service).

Possible source shown by the plugin: Cloud Computing → Chapter 2 → Cloud_Computing.pdf

8. Simple Analogy for a Presentation

Chunking is like cutting a large textbook into relevant pages.

Embedding is like converting each page into a mathematical representation of its meaning.

The vector database is like a smart library that stores those representations and quickly finds the pages most relevant to a student's question.

9. Key Message

A RAG-based Moodle plugin does not simply ask an AI model to answer a question. It first retrieves relevant information from Moodle learning materials and then provides that information to the AI model so the generated response is grounded in the actual course content.